# FIPS-Aligned STARKs with Concrete Multi-Level Post-Quantum Security

Stephen A Holmes

University of Surrey
s.a.holmes@surrey.ac.uk

**Abstract.** In regulated environments requiring FIPS 140-3 certification, production FRI-based STARKs are precluded by their reliance on algebraic hashes (e.g., Poseidon) on the security-critical path. We present a FIPS-aligned STARK architecture whose security decomposes into two independently auditable components: the information–theoretic soundness of FRI and the collision resistance of SHA-3.
Our DEEP-ALI merge compiles all AIR constraints into a single composition polynomial before folding, reducing Merkle authentication paths per query from $\Theta(\sum_\ell m_\ell)$ to $L+1$ and enabling a modular QROM decomposition $\varepsilon \le \varepsilon_{\mathrm{IT}} + \varepsilon_{\mathrm{bind}} + \varepsilon_{\mathrm{FS}}$, with each term tunable via query count, collision entropy, and extension degree. Restricting to binary folding—the only unconditionally sound regime in the literature—we derive concrete parameter sets achieving NIST post-quantum Levels 1 and 3 without conjectures, and Level 5 subject to a hard FIPS 202 "binding wall" at $q \le 2^{83}$, where the QROM term $\varepsilon_{\mathrm{bind}} = O(q^3/2^n)$ is capped by Keccak-1600's 512-bit sponge capacity regardless of output length.
Our dual-hash design uses Poseidon for internal prover acceleration while confining the verifier path to SHA-3. An open-source implementation yields Goldilocks-based proofs of $\approx$1.1–4.0 MiB, verified in 10–44 ms on AVX512 servers.

**Keywords:** STARKs · FRI · SHA-3 · FIPS compliance · post-quantum security · NIST levels · QROM · DEEP-ALI

## 1 Introduction

Scalable Transparent Arguments of Knowledge (STARKs) [1] are among the few deployed succinct proof systems requiring no trusted setup. Their post-quantum security rests on two pillars: information–theoretic soundness of the FRI (Fast Reed-Solomon Interactive Oracle Proof of Proximity) test [2] and collision resistance of the Merkle commitment hash. In principle these pillars are cleanly separated; in practice, production systems face an uncomfortable choice. Plonky2 [18], Plonky3 [19], and STWO [16] place an algebraic hash (Poseidon, Rescue Prime) on the Merkle path, capping security at the algebraic hash's margin (often an unstated $\sim$80 bits), introducing an assumption with no mature post-quantum analysis [14,15], and precluding FIPS 140-3 certification. Winterfell [20] avoids algebraic hashes by offering BLAKE3 and SHA3-256, but retains the

standard FRI topology in which each layer opens $\Theta(m)$ field elements; the resulting 32-byte digests inflate proofs and verification time, discouraging adoption of FIPS-approved hashes. Neither path yields a system that is simultaneously compact, QROM-secure, and FIPS-aligned.

**This paper** presents a STARK architecture whose sole computational assumption is collision resistance of a single FIPS 202 hash function (SHA-3 family: SHA3-256, SHA3-384, or SHA3-512, matched to the target level); low-degree-test soundness is unconditional [4]. We derive concrete parameter sets for NIST post-quantum Levels 1, 3, and—with identification of a hard limit—5. The construction rests on two mutually reinforcing techniques and a parameterised security analysis.

**Contributions.**

*(1) DEEP-ALI merge with constant authentication paths.* We embed the Algebraic Linking Identity into the FRI quotient so the verifier opens exactly $L{+}1$ Merkle leaves per query, independent of the folding arity (within the binary regime we analyse) (Theorem 3). For our recommended schedule this cuts paths from 33 to 17, making full-length SHA-3 digests practical.

*(2) Modular security decomposition.* QROM soundness factors as $\varepsilon \le \varepsilon_{\mathrm{IT}} + \varepsilon_{\mathrm{bind}} + \varepsilon_{\mathrm{FS}}$ (Theorem 4), governed by query count $r$, effective collision entropy $n$, and extension degree $e$ respectively; the triple $(r, n, e)$ tunes independently to any target $\lambda$.

*(3) NIST-compliant parameter sets.* Using binary folding—the only regime with an unconditional FRI contraction factor [4]—we derive concrete $(e, n, r)$ triples for Levels 1 ($\ge$ 128 b), 3 ($\ge$ 192 b), and 5 ($\ge$ 256 b, $q \le 2^{83}$) across quantum budgets $q \in \{2^{40}, 2^{65}, 2^{90}\}$ (Table 2). All bounds are unconditional.

*(4) The binding wall and FIPS 202 hardness.* The QROM collision bound creates a sharp threshold $q_{\max}$ beyond which the hash is insufficient (Theorem 7). Every FIPS 202 function respects an $n = 512$ wall: SHAKE256 has capacity $c = 512$ bits, bounding effective collision entropy by $\min(n_{\mathrm{out}}, c) \le 512$ (Corollary 2).

*(5) Explicit QROM bit-security accounting.* Instantiating the DFMS multi-round Fiat–Shamir framework [11] for the $K = L{+}2$ round compiled protocol (Theorem 5) reveals a $\lceil 2\log_2 K \rceil$-bit grinding penalty controlled by the folding schedule—a dependence invisible in the classical ROM.

*(6) Implementation and benchmarks.* A dual-hash architecture (Poseidon internally, SHA-3 on the verifier path) over Goldilocks with $\mathbb{F}_{p^6}$ and $\mathbb{F}_{p^8}$ tower arithmetic yields 1.1–4.0 MiB proofs verified in 10–44 ms. Byte-identical sizes across three AIR instances ($w \in \{2, 8, 16\}$, $d_c = 2$) confirm constraint-system independence. These sizes exceed algebraic-hash systems but are acceptable for regulated settings where FIPS certification is mandatory.

## 2 Preliminaries

**Goldilocks field.** $\mathbb{F}_p$ with $p = 2^{64} - 2^{32} + 1$, chosen for efficient 64-bit arithmetic and high two-adicity ($2^{32} \mid p{-}1$).

**Extension towers.** For extension degree $e$, $|\mathbb{F}_{p^e}| = p^e \approx 2^{64e}$. We construct: $\mathbb{F}_{p^3} = \mathbb{F}_p[x]/(x^3 - x - 1)$; $\mathbb{F}_{p^6} = \mathbb{F}_{p^3}[y]/(y^2 - \alpha)$ for non-square $\alpha \in \mathbb{F}_{p^3}$; $\mathbb{F}_{p^8} =$

$\mathbb{F}_{p^2}[y]/(y^4 - \beta)$ for non-fourth-power $\beta \in \mathbb{F}_{p^2}$, where $\mathbb{F}_{p^2} = \mathbb{F}_p[x]/(x^2 - 7)$. Challenge spaces have cardinality $2^{192}$, $2^{384}$, and $2^{512}$ respectively. The choice of $e$ directly controls the Fiat–Shamir ceiling relative to the target security parameter $\lambda$ (defined below); we show in Section 5 that $\mathbb{F}_{p^3}$ is insufficient for Level 1, $\mathbb{F}_{p^6}$ suffices for Levels 1–3, and $\mathbb{F}_{p^8}$ is needed for Level 5.

**Reed–Solomon codes and FRI.** Let $T$ be the trace length and $d_c$ the maximum individual constraint degree, giving composed-polynomial degree $D = d_c \cdot T$ and post-DEEP (Domain Extension for Eliminating Pretenders) target degree $d_0 = D - 1$. Then $\mathsf{RS}_{H_0}(d_0)$ is the Reed–Solomon code of degree-$\leq d_0$ polynomials evaluated on $H_0 \subset \mathbb{F}_p$, with $|H_0| = D/\rho_0$ and rate $\rho_0 = 1/32$. We use binary folding arity ($m_\ell = 2$) over $L = 16$ layers, yielding $|H_L| = |H_0|/2^L$ and compiled round count $K = L + 2 = 18$. (Corollary 1) sets $T = 2^{22}$, $d_c = 4$ ($|H_0| = 2^{29}$, $|H_L| = 2^{13}$); benchmarks (Section 10) set $T = 2^{16}$, $d_c = 2$ ($|H_0| = 2^{22}$, $|H_L| = 2^6$). For functions $f, g : S \to \mathbb{F}$, $\Delta(f, g) = |\{s : f(s) \neq g(s)\}|/|S|$; $\mathrm{dist}(f, \mathcal{C}) = \min_{g \in \mathcal{C}} \Delta(f, g)$.

**Binary folding arity.** We restrict to binary folding ($m_\ell = 2$ for every layer $\ell$). The BCIKS proximity-gap theorem [4] accommodates any arity, but the *correlated-agreement* step stitching per-coset checks into a global codeword guarantee across FRI rounds is proven only for $m = 2$, where each coset is a pair of points uniquely determining a line [4]. For $m > 2$, one must further show that the degree-$(m-1)$ polynomials fitted to individual cosets agree with a single global polynomial—an assumption unproven in the published literature [1,5], requiring either a conjectural extension of correlated agreement or a different proximity-gap argument. Since our goal is a *fully unconditional* security statement aligned with NIST post-quantum levels, every claim in Sections 3–5 rests on proven bounds with explicit contraction constants.

**SHA-3 family and sponge capacity.** FIPS 202 [21] specifies hash functions built on the Keccak-1600 permutation ($b = 1600$ bits) with capacity $c = b - r_{\mathrm{rate}}$: SHA3-256 has $c = 512$; SHA3-384 has $c = 768$; SHA3-512 has $c = 1024$; SHAKE256 has $c = 512$. The *effective collision entropy* is $n = \min(n_{\mathrm{out}}, c)$, where $n_{\mathrm{out}}$ is the output length. For the fixed-output variants $n_{\mathrm{out}} \leq c$, so $n = n_{\mathrm{out}}$. This distinction becomes critical in Section 7.

**Collision resistance.** Throughout, $\lambda$ denotes the *target security parameter* in bits, aligned with NIST post-quantum levels: $\lambda \in \{128, 192, 256\}$ for Levels 1, 3, and 5 respectively. The system achieves $\lambda$-bit security when every error term—information–theoretic and computational—is at most $2^{-\lambda}$. A hash family $\mathcal{H}$ is $(\varepsilon_{\mathrm{col}}, t)$-collision resistant if no $t$-time adversary finds a collision with probability exceeding $\varepsilon_{\mathrm{col}}$; in the post-quantum setting the adversary makes $q$ quantum queries and we write $\varepsilon_{\mathrm{col}}^{\mathrm{pq}}$ for the corresponding bound. In the QROM, $\varepsilon_{\mathrm{col}}^{\mathrm{pq}} \leq c_b\, q^3/2^n$, where $c_b \geq 1$ is the absolute constant from Zhandry's compressed-oracle bound [8]; we set $c_b = 1$ throughout. This parameterisation underpins our modular argument: the IT component depends only on FRI, the computational component only on the instantiated $H \in \mathcal{H}$, and each must remain below $2^{-\lambda}$ at the chosen level.

## 3 The DEEP-ALI Merge

The merge compiles all AIR constraints into a single univariate proximity target whose distance from the relevant Reed–Solomon code is constant whenever the instance is invalid, independently of the folding schedule. This structural transformation is what makes full-length SHA-3 digests practical: without it, binary arity over 16 layers would require $16 \times 2 + 1 = 33$ authentication paths per query; with it, the count is 17.

### 3.1 Construction

Let $\Phi_1(X), \ldots, \Phi_s(X)$ denote the individual constraint polynomials for an AIR instance over $H$. The Fiat–Shamir transcript supplies combination coefficients $\alpha_1, \ldots, \alpha_s \in \mathbb{F}_{p^e}$, and the composed constraint polynomial is $\Phi(X) = \sum_{j=1}^{s} \alpha_j \, \Phi_j(X)$. The instance is valid iff $Z_H(X) \mid \Phi_j(X)$ for every $j$. Standard STARK constructions compose constraints via a random linear combination that is committed and verified separately from the folded trace tables undergoing FRI proximity testing. The merge restructures the composition so that the constraint polynomial *is* the FRI folding target, enabling constant authentication-path verification.

The transcript additionally supplies a DEEP point $z \in \mathbb{F}_{p^e} \setminus H$, a blinding coefficient $\beta \in \mathbb{F}_{p^e}^{\times}$, and a low-degree blinding polynomial $R(X)$ with $\deg R \leq d_0$. The *DEEP-ALI merged polynomial* is

$$C(X) \;=\; \frac{\Phi(X) - \frac{\Phi(z)}{Z_H(z)} \cdot Z_H(X)}{X - z} \;+\; \beta \cdot R(X). \tag{1}$$

The layer-0 evaluation table is $f_0 = C|_{H_0}$, submitted to FRI folding. No separate constraint polynomial exists: fold consistency and constraint satisfaction are encoded jointly.

At each folding layer $\ell$ with arity $m_\ell$, let $\alpha_\ell$ denote the folding coefficient, $z_\ell \in \mathbb{F}_{p^e} \setminus H_\ell$ the layer DEEP point, and $\beta_\ell \in \mathbb{F}_{p^e}^{\times}$ a randomiser. The composition polynomial is

$$\mathsf{CP}_\ell(X) \;=\; \frac{f_\ell(X) - \sum_{j=0}^{m_\ell - 1} \alpha_\ell^j \cdot f_{\ell+1}(X^{m_\ell}) \cdot L_{\ell,j}(X)}{X^{m_\ell} - 1} \;+\; \beta_\ell \cdot \frac{f_\ell(X) - f_\ell(z_\ell)}{X - z_\ell} \tag{2}$$

where $L_{\ell,j}$ are Lagrange basis polynomials. In our construction, the prover commits $(f_\ell, \mathsf{CP}_\ell)$ in a single Merkle leaf per layer.

### 3.2 Soundness of the Merge

**Theorem 1 (DEEP-ALI degree bound and initial distance).** *Let $\mathcal{A}$ be an AIR instance with $s$ constraints of individual degree at most $d_c$ over a trace of length $T$, yielding composed-polynomial degree $D = d_c \cdot T$. Write $\rho_0 = D/|H_0|$ for the initial FRI rate and let $|\mathbb{F}_{p^e}|$ denote the extension-field size. Let $f_0 = C|_{H_0}$ with DEEP challenge $z \in \mathbb{F}_{p^e} \setminus H_0$. Define the ALI soundness error*

$$\varepsilon_{\mathrm{ali}} \;=\; \frac{s \;+\; D \;+\; 1}{|\mathbb{F}_{p^e}| - |H_0|}\,. \tag{3}$$

*(i)* Completeness: *if the AIR is valid, then* $\deg C \le d_0$ *and* $f_0 \in \mathsf{RS}_{H_0}(d_0)$.
*(ii)* Soundness: *if the AIR is invalid, then except with probability at most* $\varepsilon_{\mathrm{ali}}$,

$$\mathrm{dist}\big(f_0,\ \mathsf{RS}_{H_0}(d_0)\big)\ \ge\ 1-\rho_0-\varepsilon_{\mathrm{ali}}\,.$$

*Proof. Completeness.* When $Z_H \mid \Phi$, write $\Phi = Q \cdot Z_H$. The numerator of (1) becomes $(Q(X)-Q(z)) \cdot Z_H(X)$, which vanishes at $X = z$; division by $(X - z)$ is exact, yielding $\deg C \le d_0$. Because $H \subset H_0$ and $C$ has degree at most $d_0$, the restriction $f_0 = C|_{H_0}$ is a codeword of $\mathsf{RS}_{H_0}(d_0)$.

*Soundness.* Assume the AIR is invalid. We identify two bad events over the verifier's randomness and apply a union bound.

*Event* $E_1$ *(combination cancellation).* Because the AIR is invalid, there exist $j^\star \in [s]$ and $i^\star \in [T]$ with $\Phi_{j^\star}(\omega_T^{i^\star}) \neq 0$. The map $\alpha_{j^\star} \mapsto \Phi(\omega_T^{i^\star}) = \alpha_{j^\star}\,\Phi_{j^\star}(\omega_T^{i^\star}) + \sum_{j\neq j^\star} \alpha_j\,\Phi_j(\omega_T^{i^\star})$ is a non-trivial affine function of $\alpha_{j^\star}$. By Schwartz–Zippel, $\Pr[\Phi(\omega_T^{i^\star}) = 0] \le 1/|\mathbb{F}_{p^e}|$. A union bound over all $s$ constraint indices gives $\Pr[E_1] \le s/|\mathbb{F}_{p^e}|$.

*Event* $E_2$ *(DEEP point degeneracy).* Conditioned on $\neg E_1$, $Z_H \nmid \Phi$ and the remainder $\Phi \bmod Z_H$ is a non-zero polynomial of degree less than $T$. Define the DEEP quotient

$$D_z(X)\ =\ \frac{\Phi(X) - \frac{\Phi(z)}{Z_H(z)}\,Z_H(X)}{X-z}\,.$$

The numerator is a polynomial of degree $\deg \Phi \le D$ that vanishes at $X = z$, so the quotient $D_z$ is a polynomial of degree at most $D - 1$. Viewing the restriction $f_0 = D_z|_{H_0}$ as a function of the challenge $z$, each coefficient of its $H_0$-interpolation above degree $d_0$ is a rational function of $z$ whose numerator has degree at most $D$ after clearing the $Z_H(z)$ denominator. Because $Z_H \nmid \Phi$, at least one such numerator is not identically zero—otherwise $C$ would have degree $\le d_0$ for every $z$, contradicting $Z_H \nmid \Phi$. Hence $f_0 \in \mathsf{RS}_{H_0}(d_0)$ only when $z$ roots a nonzero univariate polynomial of degree at most $D$, giving $\Pr[E_2 \mid \neg E_1] \le (D+1)/(|\mathbb{F}_{p^e}| - |H_0|)$.

*Union bound.* $\varepsilon_{\mathrm{ali}} = \Pr[E_1] + \Pr[E_2] \le s/|\mathbb{F}_{p^e}| + (D+1)/(|\mathbb{F}_{p^e}| - |H_0|) \le (s + D + 1)/(|\mathbb{F}_{p^e}| - |H_0|)$. Conditioned on $\neg E_1 \wedge \neg E_2$, the function $f_0 = C|_{H_0}$ satisfies $\mathrm{dist}(f_0, \mathsf{RS}_{H_0}(d_0)) \ge 1 - \rho_0 - \varepsilon_{\mathrm{ali}}$.

**Corollary 1 (Concrete instantiation).** *Set* $s = 2^{10}$ *boundary constraints, trace length* $T = 2^{22}$*, constraint degree* $d_c = 4$ *(so* $D = d_c \cdot T = 2^{24}$*), evaluation domain* $|H_0| = 2^{29}$ *(rate* $\rho_0 = 1/32$*), and extension field* $|\mathbb{F}_{p^e}| = 2^{192}$ *(Goldilocks sextic extension). These values are collected in Table 2. Then Theorem 1 and* (3) *give*

$$\varepsilon_{\mathrm{ali}}\ =\ \frac{2^{10} + 2^{24} + 1}{2^{192} - 2^{29}}\ <\ 2^{-167}\,,$$

*and the initial distance satisfies*

$$\mathrm{dist}\big(f_0,\ \mathsf{RS}_{H_0}(d_0)\big)\ \ge\ 1 - \tfrac{1}{32} - 2^{-167}\ >\ 0.968\,.$$

*Partial constraint satisfaction cannot reduce this distance: a trace violating even a single AIR constraint produces the full gap, because Event* $E_1$ *in the proof of*

*Theorem 1 requires only a single non-zero constraint polynomial for the Schwartz–Zippel bound.*

### 3.3 Distance Propagation

**Theorem 2 (Proximity-gap floor—unconditional).** *Let $\sigma_\ell$ be the per-query rejection probability at layer $\ell$. For binary folding with $\rho_\ell \le 1/4$,*

$$\sigma_\ell \;\ge\; 1 - \sqrt{\rho_0}, \qquad 0 \le \ell \le L,$$

*except with probability $L \cdot O(D/|\mathbb{F}_{p^e}|)$.*

*Proof.* Binary folding gives size-2 cosets, so correlated agreement [4, Thm. 1.3] applies unconditionally. The initial distance $\delta_0 := \mathrm{dist}(f_0, \mathsf{RS}_{H_0}(d_0)) > 0.968$ (Corollary 1) exceeds the Johnson bound $1 - \sqrt{\rho_0} \approx 0.823$, placing $f_0$ in the far regime of [4, Thm. 1.2]. Induction on $\ell$: fold-consistent layers propagate the floor by correlated agreement; deviating layers are detected via $\mathsf{CP}_\ell$ (Lemma 2). The error unions over $L$ layers.

With $\rho_0 = 1/32$: $\sigma_0 \ge 1 - \sqrt{1/32} \approx 0.823$, yielding $\log_2(1/(1-\sigma_0)) = \frac{1}{2}\log_2(1/\rho_0) = 2.5$ bits per query.

### 3.4 Constant Authentication-Path Count

**Theorem 3 (Constant authentication paths).** *With the DEEP-ALI merge and combined-layer commitments, each verifier query opens exactly $L+1$ Merkle leaves.*

**Lemma 1 (Single leaf per layer).** *At layer $\ell$, the merged verification predicate for query position $i_\ell$ depends only on the leaf $(f_\ell(i_\ell), \mathsf{CP}_\ell(i_\ell))$, the next-layer value $f_{\ell+1}(i_\ell^2)$, and public data.*

*Proof.* The fold and DEEP quotients in (2) are packaged into $\mathsf{CP}_\ell(i_\ell)$; the coset sibling $f_\ell(-i_\ell)$ that conventional FRI would open is absorbed into $\mathsf{CP}_\ell$ and reconstructible from it. The verifier checks a polynomial identity relating $f_\ell(i_\ell)$, $\mathsf{CP}_\ell(i_\ell)$, $f_{\ell+1}(i_\ell^2)$, and the public challenges.

**Lemma 2 (Composition polynomial distance).** *If fold consistency is violated at any coset of layer $\ell$, then except with probability $O(1/|\mathbb{F}_{p^e}|)$,*

$$\mathrm{dist}(\mathsf{CP}_\ell|_{H_\ell},\, \mathsf{RS}_{H_\ell}(d_\ell - 1)) \;\ge\; \frac{2}{|H_\ell|} = \frac{1}{|H_{\ell+1}|}.$$

*Proof.* A violated coset produces a non-zero fold residual, so the fold-quotient term in (2) has a pole on $\{x_0, -x_0\}$ that the DEEP quotient (regular on $H_\ell$) cannot cancel. Since any degree-$(d_\ell - 1)$ polynomial is regular on the violating coset, it disagrees with $\mathsf{CP}_\ell$ at both coset points, giving the stated distance.

*Proof (Proof of Theorem 3).* Lemma 1 gives one leaf per layer plus the terminal layer-$L$ leaf: $L+1$ total. Lemma 2 makes any fold inconsistency detectable by FRI, preserving per-query soundness.

### 3.5 Public-Parameter Binding

Every STARK verifier must know which parameter set governs the proof it checks. We bind this choice into the Fiat–Shamir transcript.

**Definition 1 (Parameter identifier).** *The* public-parameter vector *is*

$$\mathsf{pp} \;=\; (p,\; e,\; \mathsf{H_{id}},\; n_{\mathrm{out}},\; L,\; (m_\ell)_{\ell=0}^{L-1},\; r,\; T,\; d_c,\; \rho_0),$$

*serialised in a fixed canonical encoding. The Fiat–Shamir sponge absorbs* $\mathsf{pp}$ *as the zeroth message, before the prover's first commitment* $\mathsf{root}_0$.

Since every subsequent challenge depends on $\mathsf{pp}$, a proof generated under one parameter set cannot verify under another absent a collision in $H$. This closes a formal requirement of the DFMS multi-round lemma [11], which assumes each round's challenge distribution is fully determined by the preceding transcript; without $\mathsf{pp}$ in the prefix, $\varepsilon_{\mathrm{FS}}$ in Theorem 4 would require an additional union bound over valid parameter sets. The binding also acts as a domain separator, preventing cross-level replay: a Level 1 proof ($e$=6, $r$=54, `SHA3-256`) and a Level 3 proof ($e$=6, $r$=79, `SHA3-384`) absorb distinct $\mathsf{pp}$ vectors, so their challenges are independently distributed even for identical AIR inputs. Because $\mathsf{pp}$ is the first sponge input and is echoed in the proof-artifact header, verifiers can identify parameters before processing any prover-supplied data, matching the module-boundary separation expected by FIPS 140-3.

## 4 Modular Security Decomposition

**Theorem 4 (Modular security decomposition).** *Let $\Pi$ be the DEEP-ALI-FRI STARK over $\mathbb{F}_{p^e}$ compiled via Fiat–Shamir with hash $H$ having effective collision entropy $n = \min(n_{\mathrm{out}}, c)$ bits. For any $(q,t)$-QROM adversary $\mathcal{A}$:*

$$\varepsilon_{\Pi}(q) \;\le\; \underbrace{(1-\sigma_0)^r + \frac{r\,(s+D+L+1)}{|\mathbb{F}_{p^e}| - |H_0|}}_{\varepsilon_{\mathrm{IT}}} + \underbrace{(L{+}2)\,c_b\frac{q^3}{2^n}}_{\varepsilon_{\mathrm{bind}}} + \underbrace{\frac{K^2(q{+}1)^2}{|\mathbb{F}_{p^e}|}}_{\varepsilon_{\mathrm{FS}}} \qquad (4)$$

*Proof.* The argument proceeds through five hybrids. Let $\mathcal{A}$ be any $t$-time adversary producing a proof for an invalid statement.

*Hybrid 0 (Real system).* The adversary interacts with the real compiled system using $H$ for all operations.

*Hybrid 1 (Ideal commitments).* Replace Merkle commitments with ideal ones. Any distinguishing advantage requires a collision in $H$. QROM collision resistance [8] with a union bound over $L+2$ trees gives $\varepsilon_{\mathrm{bind}} \le (L{+}2)\,c_b\,q^3/2^n$.

*Hybrid 2 (DEEP conditioning).* Condition on all DEEP challenges (from $\mathbb{F}_{p^e}$) avoiding bad events. For each of the $r$ queries the layer-0 DEEP challenge incurs an ALI error of $\varepsilon_{\mathrm{ali}} = (s+D+1)/(|\mathbb{F}_{p^e}| - |H_0|)$ by Theorem 1, and each of the $L$ fold-layer DEEP points adds at most $1/|\mathbb{F}_{p^e}|$. A union bound over all $r$ queries gives $\Pr[\mathrm{bad}] \le r\,(s+D+L+1)/(|\mathbb{F}_{p^e}| - |H_0|)$.

*Hybrid 3 (IT soundness).* The adversary must fool FRI proximity testing. By Theorem 2, each query rejects with probability $\sigma_0$. For $r$ independent queries: $\Pr[\text{accept}] \le (1-\sigma_0)^r$. The Theorem 2 exception probability $L \cdot O(D/|\mathbb{F}_{p^e}|)$ is dominated by the DEEP-conditioning term in Hybrid 2 and is therefore already accounted for in $\varepsilon_{\mathrm{IT}}$.

*Hybrid 4 (Fiat–Shamir).* Apply the DFMS multi-round measure-and-reprogram lemma [11] to the $K$-round transcript with minimum challenge space $|\mathbb{F}_{p^e}|$. The lemma adds $K^2(q+1)^2/|\mathbb{F}_{p^e}|$ independently of the IT error.

Combining all terms gives (4).

### 4.1 The Three Ceilings

In bits ($\kappa = -\log_2 \varepsilon$), the three terms give:

$$\kappa_{\mathrm{IT}}(r) = r \cdot \log_2 \tfrac{1}{1-\sigma_0} = \tfrac{r}{2} \log_2 \tfrac{1}{\rho_0} \approx 2.5\, r \tag{5}$$

$$\kappa_{\mathrm{bind}}(n,q) = n - 3\log_2 q - \log_2((L{+}2)\, c_b) \approx n - 3\log_2 q - 4.2 \tag{6}$$

$$\kappa_{\mathrm{FS}}(e,q) = 64e - 2\log_2 K - 2\log_2 q \approx 64e - 8.3 - 2\log_2 q \tag{7}$$

with $\kappa_{\mathrm{sys}} = \min(\kappa_{\mathrm{IT}}, \kappa_{\mathrm{bind}}, \kappa_{\mathrm{FS}})$. The three parameters $(r, n, e)$ can be tuned independently. The DEEP-ALI merge is the structural enabler: without it, increasing $n$ (larger digests) would inflate proof size proportionally to $\sum_\ell m_\ell$.

### 4.2 Fiat–Shamir Ceiling Dictates Extension Degree

Table 1 confirms that $\mathbb{F}_{p^3}$ falls below Level 1 even at $q = 2^{32}$ (the $\lceil 2\log_2 K \rceil \approx 8.3$-bit DFMS penalty consumes too large a share of the 192-bit challenge space), $\mathbb{F}_{p^6}$ suffices for Levels 1–3, and $\mathbb{F}_{p^8}$ is required for Level 5.

**Table 1.** Fiat–Shamir ceiling $\kappa_{\mathrm{FS}}$ (bits) for binary arity ($K = 18$).

| | $q = 2^{32}$ | $q = 2^{40}$ | $q = 2^{65}$ | $q = 2^{90}$ |
|---|---|---|---|---|
| $\mathbb{F}_{p^3}$ ($2^{192}$) | 120 | 104 | 54 | 4 |
| $\mathbb{F}_{p^6}$ ($2^{384}$) | 312 | 296 | 246 | 196 |
| $\mathbb{F}_{p^8}$ ($2^{512}$) | 440 | 424 | 374 | 324 |

### 4.3 QROM Soundness

**Theorem 5 (QROM soundness).** *The DEEP-ALI-FRI STARK over $\mathbb{F}_{p^e}$ with binary arity ($L = 16$, $K = 18$), rate $\rho_0 = 1/32$, effective collision entropy $n$, per-query detection $\sigma_0 \ge 1 - \sqrt{\rho_0} \approx 0.823$, $r$ queries, and quantum budget $q$ achieves*

$$\kappa_{\mathrm{sys}} = \min\bigl(2.50\, r,\ \ n - 3\log_2 q - 4.2,\ \ 64e - 8.3 - 2\log_2 q\bigr).$$

*Proof (Proof sketch).* Instantiate the decomposition of Theorem 4 with the unconditional proximity-gap floor $\sigma_0 \geq 1 - \sqrt{\rho_0}$ (Theorem 2), giving $\log_2(1/(1-\sigma_0)) = \frac{1}{2}\log_2(1/\rho_0) = 2.5$ bits per query. The five-hybrid argument—ideal commitments, DEEP conditioning, multi-round Fiat–Shamir via [11]—substitutes $|\mathbb{F}_{p^e}| = p^e$ in the FS denominator and $2^n$ with $n = \min(n_{\text{out}}, c)$ in the binding denominator. Converting each term to bit-security yields three ceilings of (5)–(7).

### 4.4 Quantum Structure and Fold-Layer Absorption

The $\lceil 2\log_2 K \rceil$-bit grinding penalty in $\kappa_{\text{FS}}$ arises because the DFMS framework treats each fold-layer commitment as an independent adaptive round. After the DEEP-ALI merge, however, every fold-layer commitment is determined by $\mathsf{root}_0$ and the public verifier challenges, reducing a quantum adversary's freedom to a superposition over choices of $\mathsf{root}_0$ alone. Whether this deterministic dependence allows the $L$ fold-layer rounds to be *absorbed* into a single effective round—eliminating the $\lceil 2\log_2 K \rceil \approx 8.3$-bit penalty—remains an open problem for all FRI-based proof systems. A positive resolution would raise effective security by $\approx \log_2(L+1) \approx 4$ bits across all parameter sets in Table 2.

### 4.5 Hash-Upgrade Independence

**Theorem 6 (Hash-upgrade independence).** *Replacing the hash $H$ with any collision-resistant $H'$ preserves all IT parameters. The computational bound becomes $\varepsilon_{H'}$; overall security becomes* $\min(-\log_2 \varepsilon_{\text{IT}}, -\log_2 \varepsilon_{H'})$.

*Proof.* By Theorem 4, the IT and computational components are additive and independent. The IT component depends only on FRI parameters; the computational component depends only on $\varepsilon_H$. Replacing $H$ with $H'$ changes the latter.

For instance, substituting `SHA3-384` gives $\lfloor 384/3 \rfloor = 128$ bits under the BHT model [9] with no change to $e$, $r$, or the folding schedule, providing regulatory flexibility if future NIST guidance revises the categorisation of `SHA3-256`.

## 5 Multi-Level Parameterisation

We combine the three ceilings to derive parameter triples $(e, n, r)$ for each NIST level. All sets use binary arity and are therefore fully unconditional. Table 2 yields the following conclusions.

Level 1 ($\geq 128$ bits) is achieved for all $q \leq 2^{90}$ with $\mathbb{F}_{p^6}$, upgrading the hash as $q$ grows; the binding term at $\kappa_{\text{bind}} = 132$ bits ($q = 2^{40}$) is the tightest constraint, yielding $\kappa_{\text{sys}} = 131$ after the union bound and floor, while the remaining Level 1 configurations are $\kappa_{\text{IT}}$-limited at $\kappa_{\text{sys}} = 134$. Level 3 ($\geq 192$ bits) holds with $\mathbb{F}_{p^6}$ and `SHA3-384`/`SHA3-512`; at $q = 2^{90}$ the FS ceiling ($\kappa_{\text{FS}} = 196$) becomes tight against $\kappa_{\text{IT}} = 197.5$ with a gap of only 1.8 bits, giving $\kappa_{\text{sys}} = 195$, so deployments requiring margin should use $\mathbb{F}_{p^8}$. Level 5 ($\geq 256$ bits) requires $\mathbb{F}_{p^8}$ and `SHA3-512`; the binding wall is reached at $q \approx 2^{83}$ ($\kappa_{\text{bind}} \approx 256$) and breached at $q = 2^{90}$ ($\kappa_{\text{bind}} = 238$, $\kappa_{\text{sys}} = 237$), below the NIST-5 floor.

**Table 2.** Recommended NIST-compliant parameter sets. All entries unconditional (binary arity, proven contraction). $\kappa_{\mathrm{sys}} := \lfloor -\log_2(2^{-\kappa_{\mathrm{IT}}} + 2^{-\kappa_{\mathrm{bind}}} + 2^{-\kappa_{\mathrm{FS}}}) \rfloor$, computed from pre-rounded $\kappa$ values. †: binding wall violated.

| Level | $q$ | Ext. | Hash | $r$ | $\kappa_{\mathrm{IT}}$ | $\kappa_{\mathrm{bind}}$ | $\kappa_{\mathrm{FS}}$ | $\kappa_{\mathrm{sys}}$ |
|---|---|---|---|---|---|---|---|---|
| 1 | $2^{40}$ | $\mathbb{F}_{p^6}$ | SHA3-256 | 54 | 135 | 132 | 296 | **131** |
| | $2^{65}$ | $\mathbb{F}_{p^6}$ | SHA3-384 | 54 | 135 | 185 | 246 | **134** |
| | $2^{90}$ | $\mathbb{F}_{p^6}$ | SHA3-512 | 54 | 135 | 238 | 196 | **134** |
| 3 | $2^{40}$ | $\mathbb{F}_{p^6}$ | SHA3-384 | 79 | 197.5 | 260 | 296 | **197** |
| | $2^{65}$ | $\mathbb{F}_{p^6}$ | SHA3-512 | 79 | 197.5 | 313 | 246 | **197** |
| | $2^{90}$ | $\mathbb{F}_{p^6}$ | SHA3-512 | 79 | 197.5 | 238 | 196 | **195** |
| 5 | $2^{40}$ | $\mathbb{F}_{p^8}$ | SHA3-512 | 105 | 262.5 | 388 | 424 | **262** |
| | $2^{65}$ | $\mathbb{F}_{p^8}$ | SHA3-512 | 105 | 262.5 | 313 | 374 | **262** |
| | $2^{90}$ | $\mathbb{F}_{p^8}$ | SHA3-512 | 105 | 262.5 | 238 | 324 | **237**$^{\dagger}$ |

## 6 The Binding Wall

**Theorem 7 (Binding wall).** *For target security level $\lambda$ and effective collision entropy $n$ bits, the maximum QROM quantum budget at which the Merkle binding term sustains $\lambda$-bit security is*

$$q_{\max} = 2^{\lfloor (n-\lambda-\lceil \log_2((L+2)\, c_b)\rceil)/3 \rfloor}. \tag{8}$$

*Proof.* Require $\kappa_{\mathrm{bind}} \geq \lambda$: $n - 3\log_2 q - \lceil \log_2((L+2)\, c_b)\rceil \geq \lambda$. Solving for $q$ gives the result.

**Table 3.** Maximum quantum budget $q_{\max}$ before the binding wall is breached. An entry of $<1$ indicates the hash cannot meet the target level for any $q$.

| Hash | Level 1 ($\lambda$=128) | Level 3 ($\lambda$=192) | Level 5 ($\lambda$=256) |
|---|---|---|---|
| SHA3-256 ($n = 256$) | $2^{41}$ | $2^{19}$ | $<1$ |
| SHA3-384 ($n = 384$) | $2^{83}$ | $2^{62}$ | $2^{41}$ |
| SHA3-512 ($n = 512$) | $2^{126}$ | $2^{105}$ | $2^{83}$ |

The wall is not specific to our construction: it applies to any Merkle-committed proof system whose QROM security reduces to hash collision resistance. The $O(q^3/2^n)$ bound is tight for unstructured hash functions [8].

## 7 The FIPS-202 Capacity Limit

We show that the binding wall cannot be circumvented within FIPS 202 by using SHAKE256 with extended output $n_{\mathrm{out}} > 512$.

**Theorem 8 (Sponge collision bound).** *Let $\mathcal{H}$ be a sponge with capacity $c$ and output $n_{\text{out}}$, and set $n_{\text{eff}} := \min(n_{\text{out}}, c)$. Classically, collision resistance is at most $n_{\text{eff}}/2$ bits. In the QROM, a $q$-query adversary satisfies*

$$\varepsilon_{\text{col}}^{\text{pq}} \;\leq\; O\big(q^3/2^{n_{\text{eff}}}\big)\,,$$

*and the bound is tight when $n_{\text{out}} > c$.*

*Proof.* For $n_{\text{out}} \leq c$, sponge indifferentiability [7] reduces collision finding to a random oracle on $n_{\text{out}}$-bit outputs; Zhandry's compressed-oracle bound [8] then gives $O(q^3/2^{n_{\text{out}}})$. For $n_{\text{out}} > c$, any two inputs whose $c$-bit capacity states collide produce identical outputs of arbitrary length, so the same bound applies with $n_{\text{out}}$ replaced by $c$; tightness follows from generic collision search on the $c$-bit internal state [9].

**Table 4.** FIPS 202 functions and effective collision entropy $n_{\text{eff}} = \min(n_{\text{out}}, c)$. An entry of $<1$ indicates the function cannot meet Level 5 for any $q$.

| Function | $r_{\text{rate}}$ | $c$ | $n_{\text{out}}$ | $n_{\text{eff}}$ | $q_{\max}$ (L5) |
|---|---|---|---|---|---|
| `SHA3-256` | 1088 | 512 | 256 | 256 | $<1$ |
| `SHA3-384` | 832 | 768 | 384 | 384 | $2^{41}$ |
| `SHA3-512` | 576 | 1024 | 512 | 512 | $2^{83}$ |
| `SHAKE128` | 1344 | 256 | var. | 256 | $<1$ |
| `SHAKE256` | 1088 | 512 | var. | 512 | $2^{83}$ |

Table 4 reveals the critical asymmetry: SHAKE256 has capacity $c = 512$, identical to SHA3-256 and half that of SHA3-512. Extended output does not help: $n_{\text{eff}} = \min(n_{\text{out}}, 512) = 512$ for any $n_{\text{out}} \geq 512$.

**Corollary 2 (FIPS 202 binding limit).** *No FIPS 202 function achieves $n > 512$. The Level 5 binding wall at $q_{\max} = 2^{83}$ cannot be raised by any choice of function or output length within the standard.*

*Proof.* From Table 4, the maximum $n_{\text{eff}}$ is 512 (SHA3-512); SHAKE256 variants achieve at most $n_{\text{eff}} = 512$ by Theorem 8. Substituting $n = 512$ into Theorem 7: $q_{\max} = 2^{\lfloor (512-256-5)/3 \rfloor} = 2^{83}$.

This is not a limitation of our construction but a consequence of the collision resistance available from Keccak-1600 in the QROM: any Merkle-committed proof system targeting Level 5 with $q > 2^{83}$ would require a hash with capacity exceeding 1024 bits, or a commitment scheme whose QROM security does not reduce to collision resistance. At $q = 2^{40}$, Level 5 achieves $\kappa_{\text{sys}} = 262$ with 6 bits of margin above the 256-bit threshold—well within the most optimistic $O(2^{40})$–$O(2^{50})$ projections for fault-tolerant quantum operations [13]. The wall is a theoretical boundary, not a practical vulnerability.

## 8 Dual-Hash Architecture and FIPS Compliance

**Prover and verifier paths.** The prover builds ephemeral intermediate Merkle trees with Poseidon over $\mathbb{F}_p$ for internal bookkeeping; these never reach the verifier. Every Poseidon-committed leaf is re-committed under the appropriate SHA-3 variant before any value enters the Fiat–Shamir transcript or proof artifact. Final Merkle roots, authentication paths, and the Fiat–Shamir sponge use SHA-3 exclusively, modeled as a quantum random oracle.

**Lemma 3 (Poseidon transcript erasure).** *For any $(q,t)$-QROM adversary $\mathcal{A}$, the view $\mathcal{V}_{\mathcal{A}}$ in the dual-hash protocol is identically distributed to the view $\mathcal{V}'_{\mathcal{A}}$ in the modified protocol where every internal Poseidon invocation is replaced by a fresh uniform $\mathbb{F}_p$-element of matching output arity.*

*Proof.* Poseidon outputs appear only as inputs to SHA-3 re-commitments, so $\mathcal{A}$'s view depends on them solely through their SHA-3 images. Modeled as a QRO, SHA-3 maps distinct inputs to independent uniform outputs; substituting an independent uniform $\mathbb{F}_p$-element for any Poseidon output therefore leaves the SHA-3 output distribution unchanged. Induction over prover execution order gives the claim.

**Security separation.** Lemma 3 reduces the dual-hash protocol to one in which Poseidon is private randomness, so any QROM forgery must break SHA-3 Merkle binding ($\varepsilon_{\mathrm{bind}}$), QROM Fiat–Shamir ($\varepsilon_{\mathrm{FS}}$), or FRI soundness ($\varepsilon_{\mathrm{IT}}$), yielding Theorem 4. A Poseidon compromise affects prover performance but not security: the CMVP validation boundary covers only the SHA-3 path.

**FIPS compliance.** Confining the verifier's trust path to SHA-3 is a necessary precondition for FIPS 140-3 certification. No production STARK achieves this: Plonky2, Plonky3, and STWO place algebraic hashes on the commitment path; Winterfell supports SHA3-256 but its standard FRI topology opens $\Theta(m)$ field elements per layer, with 32-byte digests inflating proofs—a cost that leads its developers to default to Blake3 (not FIPS-approved). A FIPS-approved hash alone is insufficient: NIST post-quantum levels further require a concrete QROM soundness analysis binding parameters to bit-security targets, which no existing STARK provides.

## 9 Soundness, Zero-Knowledge, and Extractability

**Lemma 4 (Perfect HVZK).** *If $r \le d_0$, the honest verifier's view is identically distributed for any two witnesses satisfying the AIR on the same public input.*[1]

*Proof.* The prover commits to $C(X) = D_z(X) + \beta R(X)$, where $\deg D_z \le d_0$, $\beta \in \mathbb{F}_{p^e}^{\times}$ is uniform, and $R$ is uniform on $\{f \in \mathbb{F}_{p^e}[X] : \deg f \le d_0\}$ independently of the witness. Hence $\beta R$ is uniform on that space and translation by $D_z$ preserves

[1] Satisfied with large margin: all parameter sets in Table 2 have $d_0 \ge 2^{24}$ while $r \le 105$.

uniformity, so $C$ is itself uniform and witness-independent. By $(d_0+1)$-wise independence, any $r \le d_0+1$ evaluations of $C$ are jointly uniform on $\mathbb{F}_{p^e}^r$. The verifier's view—Merkle roots, $r$ openings, and Fiat–Shamir challenges—is a deterministic function of these uniform evaluations and the random oracle, hence witness-independent. Non-interactive HVZK in the (Q)ROM follows by the standard simulator argument.

**Theorem 9 (Argument of Knowledge).** *The Fiat–Shamir compiled DEEP-ALI merged STARK is an argument of knowledge in the classical ROM with knowledge error* $\varepsilon_{\mathrm{know}} \le \varepsilon_{\Pi}(q) + q^2/2^{n_{\mathrm{out}}+1}$*, where* $n_{\mathrm{out}}$ *is the oracle output length (our parameter sets have* $n_{\mathrm{out}} \ge 2\lambda$*). A straight-line extractor runs in time* $O(q + |H|)$*.*

*Proof (Proof sketch).* Given an accepting proof $\pi$ and the random-oracle transcript $\tau$, the extractor reconstructs the evaluation vector $\mathbf{t}$ by matching oracle queries in $\tau$ to Merkle authentication paths in $\pi$, in time $O(q + |H|)$. Reconstruction fails only if $\pi$ contains a valid path through an unqueried hash, which occurs with probability at most $q^2/2^{2\lambda+1}$ by the birthday bound. If the recovered trace violates any constraint, Theorem 4 bounds the acceptance probability at $\varepsilon_{\Pi}$.

*Remark 1 (QROM and splicing resistance).* Theorem 9 establishes knowledge soundness in the classical ROM; computational soundness in the QROM follows from Theorem 4. An explicit QROM knowledge-soundness bound remains open for all deployed STARKs; the CMS framework [12] with `SHA3-384` ($\lambda = 192$) offers a concrete route, as the collapsing denominator no longer dominates. We also note that *splicing*—combining components from two valid proofs for distinct statements—requires consistency with a single Merkle root; the resulting function $C' \colon H \to \mathbb{F}$ is not a valid codeword, so FRI rejects with probability $\ge 1-(1-\sigma_0)^r$ and ideal replacement costs $\varepsilon_{\mathrm{col}}^{\mathrm{pq}}(H)$.

## 10 Empirical Validation

We implement all configurations over Goldilocks with an Arkworks-based [17] codebase, extending to $\mathbb{F}_{p^6}$ and $\mathbb{F}_{p^8}$ tower arithmetic with all three SHA-3 variants. Binary arity throughout; $L = 16$, $K = 18$. Benchmarks on AWS `c5.4xlarge` (Cascade Lake, AVX-512, 16 Rayon threads) span trace lengths $T \in [2^{11}, 2^{24}]$ across degree-2 AIR instances ($w \in \{2, 8, 16\}$), with all parameters ($r$, $n$, $e$, $L$, $K$, $w$, $T$) tunable at configuration time. Source code and raw data at [27].

### 10.1 Proof Sizes and Verification

Proof sizes range from 1 081 KiB (Level 1, `SHA3-256`) to 4 067 KiB (Level 5, `SHA3-512`), driven by larger extension-field payloads ($8e$ bytes per $\mathbb{F}_{p^e}$ element), longer digests (32 → 64 bytes), and higher query counts (54 → 105). Verification remains below 44 ms and proving under 1.1 s across all levels. While larger than algebraic-hash-based systems, these sizes are acceptable for FIPS-mandated deployments (e.g., financial/government infrastructure) where certification outweighs bandwidth considerations.

**Table 5.** Proof size and timing (AWS `c5.4xlarge`, 16 vCPU, AVX-512). Binary arity, $L$=16 folding layers, $K = 18$ compiled rounds. Degree-$d_L$ final polynomial sent in full. Times averaged over three AIR instances. Prover times include dual-hash acceleration.

| Level | Ext. | Hash | $r$ | Proof (KiB) | $t_{\text{verify}}$ (ms) | $t_{\text{prove}}$ (s) |
|---|---|---|---|---|---|---|
| 1 | $\mathbb{F}_{p^6}$ | `SHA3-256` | 54 | 1 081 | 10.2 | 0.68 |
| 1 | $\mathbb{F}_{p^6}$ | `SHA3-384` | 54 | 1 573 | 13.4 | 0.71 |
| 1 | $\mathbb{F}_{p^6}$ | `SHA3-512` | 54 | 2 065 | 20.2 | 0.71 |
| 3 | $\mathbb{F}_{p^6}$ | `SHA3-384` | 79 | 2 301 | 19.6 | 0.70 |
| 3 | $\mathbb{F}_{p^6}$ | `SHA3-512` | 79 | 3 021 | 29.5 | 0.71 |
| 5 | $\mathbb{F}_{p^8}$ | `SHA3-512` | 105 | 4 067 | 43.3 | 0.98 |

### 10.2 Distance Contraction

Table 6 reports the empirical per-layer contraction $\sigma_\ell$ at the 0.1-percentile. Theorem 2 proves the unconditional floor $\sigma_\ell \geq 1 - \sqrt{\rho_0} \approx 0.823$ via the BCIKS proximity-gap theorem at the Johnson radius; the observed values instead track the list-decoding capacity $1-\rho_0 \approx 0.969$. The gap $\sqrt{\rho_0}-\rho_0 \approx 0.145$ is the standard Johnson-to-capacity slack for RS proximity testing. A capacity-regime proximity-gap theorem would raise $\kappa_{\text{IT}}$ from $2.5\,r$ to $5\,r$ bits per query (cf. STIR [23]); our parameter sets in Table 2 use the Johnson bound and are therefore conservative by this factor. The extension degree $e$ enters only the ALI sub-term of (4) and the FS term (7), which is why all three extensions yield identical $\sigma_\ell$.

**Table 6.** Empirical per-layer contraction ratios $\sigma_\ell$ (0.1-percentile over $10^6$ trials, $T = 2^{20}$, binary arity). Theorem 2 proves the unconditional floor $\sigma_\ell \geq 0.823$; the observed values exceed this by $\sim 0.15$, consistent with the bound's worst-case looseness (see §10.2).

| Extension | $\delta_0$ | $\sigma_1$ | $\sigma_2$ | $\sigma_3$ |
|---|---|---|---|---|
| $\mathbb{F}_{p^3}$ | 0.967 | 0.976 | 0.976 | 0.975 |
| $\mathbb{F}_{p^6}$ | 0.967 | 0.976 | 0.975 | 0.975 |
| $\mathbb{F}_{p^8}$ | 0.967 | 0.976 | 0.976 | 0.975 |

### 10.3 Constraint-System Independence

Theorem 3 predicts that proof size depends only on $|H_0|$, the folding schedule, and $r$—not on trace width $w$, constraint count $s$, or constraint degree $d_c$. We validate this across three AIR instances spanning the range of production VM constraint systems.

- FIB (Fibonacci): $w = 2$, $s = 1$, $d_c = 2$.
- REC (Multi-register recurrence): $w = 8$, $s = 8$, $d_c = 2$. Eight coupled bilinear recurrences representative of STARK VM register-file transitions.
- POS (Poseidon-round AIR): $w = 16$, $s = 16$, $d_c = 2$. A $16 \times 16$ MDS layer exercising wide-trace commitment; serves as a width-16 benchmark rather than a secure Poseidon instance.

**Table 7.** Constraint-system independence ($K$=18, binary arity, $L$=16). Proof sizes are byte-identical within each level. AWS `c5.4xlarge`, 16 vCPU, AVX-512.

| | | AIR params | | | Proof | Time | |
|---|---|---|---|---|---|---|---|
| Level | AIR | $w$ | $s$ | $d_c$ | (KiB) | Prove (s) | Verify (ms) |
| 1 | FIB | 2 | 1 | 2 | 1 081 | 0.69 | 10.2 |
| | REC | 8 | 8 | 2 | 1 081 | 0.67 | 10.2 |
| | POS | 16 | 16 | 2 | 1 081 | 0.70 | 10.2 |
| 3 | FIB | 2 | 1 | 2 | 2 301 | 0.72 | 19.5 |
| | REC | 8 | 8 | 2 | 2 301 | 0.69 | 19.7 |
| | POS | 16 | 16 | 2 | 2 301 | 0.68 | 19.6 |
| 5 | FIB | 2 | 1 | 2 | 4 067 | 1.00 | 43.2 |
| | REC | 8 | 8 | 2 | 4 067 | 0.97 | 43.2 |
| | POS | 16 | 16 | 2 | 4 067 | 0.99 | 43.3 |

**Table 8.** Trust-chain comparison. *FIPS-Ver.*: verifier uses only FIPS-approved primitives. *NIST Lvls*: QROM-derived parameters at listed levels. *PQ bound*: proven post-quantum bits. *Paths/q*: Merkle paths per FRI query.

| System | Hash(es) | FIPS-Ver. | NIST Lvls | PQ bound | Paths/q |
|---|---|---|---|---|---|
| STARK [2] | coll.-res. | No | — | — | $\sum m_\ell$ |
| ethSTARK [22] | Poseidon | No | — | — | $\sum m_\ell$ |
| Plonky2 [18] | Poseidon | No | — | — | $\sum m_\ell$ |
| Plonky3 [19] | config. | Partial$^\dagger$ | — | — | $\sum m_\ell$ |
| STWO [16] | Poseidon2 | No | — | — | $\sum m_\ell$ |
| STIR [23] | config. | Partial$^\dagger$ | — | — | $O(\log d)$ |
| Binius [24] | Groestl | No | — | — | $\sum m_\ell$ |
| Winterfell [20] | BLAKE3/SHA3 | Partial$^\dagger$ | — | — | $\sum m_\ell$ |
| **Ours** | **SHA-3** | **Yes ✓** | **1/3/5 ✓** | **128/195/262 b** | **L+1** |

$^\dagger$Framework permits a FIPS hash, but no published instantiation is fully FIPS-restricted or supplies NIST-level parameters.

Three observations confirm the predictions. First, proof sizes are byte-identical within each level (1 081/2 301/4 067 KiB), as the DEEP-ALI merge makes FRI folding, commitments, and query openings operate on a fixed-size domain with $L$+1 paths per query. Second, prover time varies by less than 6% across instances despite a 16× range in $s$, confirming that NTT and Merkle-hashing dominate. Third, verification time is constant within each level to within 2%, confirming that the verifier's work is entirely post-merge.

## 11 Related Work

No prior system simultaneously achieves FIPS-only cryptography, constant authentication paths per query, concrete multi-level post-quantum bounds, and a formal QROM analysis.

Ben-Sasson et al. introduced STARKs [1] and FRI [2]; DEEP-FRI [6] improved soundness via out-of-domain sampling. Our merge extends DEEP to the full ALI composition, collapsing per-query Merkle cost to a fixed $L + 1$. ethSTARK [22] introduced DEEP for large-prime STARKs using Poseidon; our merge generalises it while removing the algebraic hash dependency. Plonky2 [18] pioneered

Goldilocks with extension-field challenges but relies on Poseidon throughout. Plonky3 [19] supports configurable hashes including Keccak; when so configured it approaches a SHA-3-only design but provides no formal security decomposition, post-quantum bound, or binding-wall analysis, and without the merge still requires $\sum_\ell m_\ell$ paths per query. STWO [16] achieves high prover performance via circle-group STARKs with Poseidon2.

STIR [23] reduces FRI interaction to $O(\log d)$ queries total, addressing a complementary bottleneck to our per-query cost reduction. Composing the two is an open direction.

Binius [24] operates over binary tower fields; no concrete QROM bound has been published. Basefold [25] generalises FRI to arbitrary linear codes but provides no FIPS or post-quantum parameter set. Lattice-based arguments (LaBRADOR [26]) require structured reference strings and are complementary to the transparency we target.

Our QROM analysis builds on Zhandry's compressed-oracle framework [8] and the DFMS multi-round paradigm [11,10]. To our knowledge we are the first to carry the analysis through the full STARK stack with concrete bit-security accounting, making explicit how the FIPS 202 sponge capacity partitions between collision resistance and extraction loss.

## 12 Conclusion

We have presented a NIST-compliant STARK framework achieving Levels 1 and 3 using only FIPS-approved SHA-3 and unconditionally proven FRI soundness with binary folding arity, and Level 5 for $q \leq 2^{83}$. The three-term QROM decomposition enables independent tuning of query count $r$, effective collision entropy $n$, and extension degree $e$, mapping each NIST level and quantum budget to concrete, unconditional security guarantees. This compliance incurs roughly an order-of-magnitude proof-size premium over algebraic-hash STARKs (e.g. Plonky2's ∼200 KiB versus our 1.1–4.0 MiB), a tradeoff we regard as the price of admission in regulated sectors—finance, defence, healthcare—where a non-FIPS-certifiable prover is not deployable at any size.

The binding wall (Theorem 7) is the fundamental limitation: the QROM collision bound $O(q^3/2^n)$ caps each level's maximum quantum budget. Corollary 2 proves this is a hard limit within FIPS 202—the Keccak-1600 capacity bounds effective collision entropy to 512 bits regardless of output length, so SHAKE256 ($c = 512$) provides no improvement over SHA3-512.

Open directions include: resolving fold-layer absorption (Section 4.4) to eliminate the $\lceil 2\log_2 K \rceil$-bit DFMS penalty; extending the merge to multi-table AIR with cross-table lookups; composing with STIR [23] to reduce $K$; establishing QROM knowledge soundness via CMS [12]; and investigating hash primitives with $c > 1024$ to lift the Level 5 wall.

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